

Kubernetes Architecture : ->

<https://youtu.be/zu2le0KLJfs?si=sIZv4KJuio3hy6XH> => yt_link

Kubernetes Cluster Explained (Beginner Friendly)

What is a Kubernetes Cluster

A Kubernetes cluster is a group of machines (nodes) that work together to run containerized applications. It manages deployment, scaling, and operation of applications automatically.

Why Kubernetes is Used in DevOps

In real-world DevOps, applications need to be:

- Highly available
- Scalable
- Fault-tolerant

Kubernetes helps by:

- Automatically scaling applications based on demand
- Restarting failed containers
- Managing deployments with minimal or no downtime
- Handling communication between services

Kubernetes Architecture Overview

A Kubernetes cluster mainly consists of two parts:

- Master Node (Control Plane)
- Worker Nodes

Master Node (Control Plane)

The master node manages the entire cluster and makes decisions.

Components:

API Server

Acts as the entry point for all requests. It receives commands and communicates with other components.

etcd

A key-value store that keeps all cluster data. It acts like a database storing the current state.

Controller Manager

Ensures the system is running as expected. It monitors and fixes issues like failed pods.

Scheduler

Decides which worker node should run a particular pod based on available resources.

Worker Nodes

Worker nodes are responsible for running the applications.

Components:

kubelet

Ensures containers are running properly inside pods. It communicates with the master node.

kube-proxy

Handles networking and traffic routing between services and pods.

Pods

Smallest unit in Kubernetes. Each pod contains one or more containers.

Key Responsibilities of Kubernetes

Auto-scaling

Adjusts the number of running containers based on traffic.

Self-healing

Restarts failed containers and replaces unhealthy pods automatically.

Traffic Distribution

Balances incoming traffic across multiple containers to ensure smooth performance.

Cloud-Independent Platform

Kubernetes is platform-independent and can run on:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- Local environments such as Minikube or Kind

This makes Kubernetes highly flexible and portable across different environments.

Real-World Analogy

A Kubernetes cluster can be compared to a managed system:

- Master Node acts like a manager controlling operations
- Worker Nodes act like employees doing the actual work

- Pods are tasks assigned to workers

The system ensures:

- Work is distributed properly
- Failures are handled automatically
- Performance remains stable

Summary

Kubernetes is a powerful system that:

- Manages containerized applications
- Automates deployment and scaling
- Ensures reliability and performance
- Works across cloud and local environments

It acts as a central system that keeps everything organized and running efficiently.